Blockchain Technology



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Blockchain is the future of trust, empowering individuals and organizations to transact, collaborate, and innovate in a secure, transparent, and decentralized manner."

- Joseph Lubin, Co-founder of Ethereum





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Section 2 Ethereum Blockchain

Section 3 Blockchain in Industry : Supply Chain Management



Section 1

Blockchain

01 What is Blockchain?

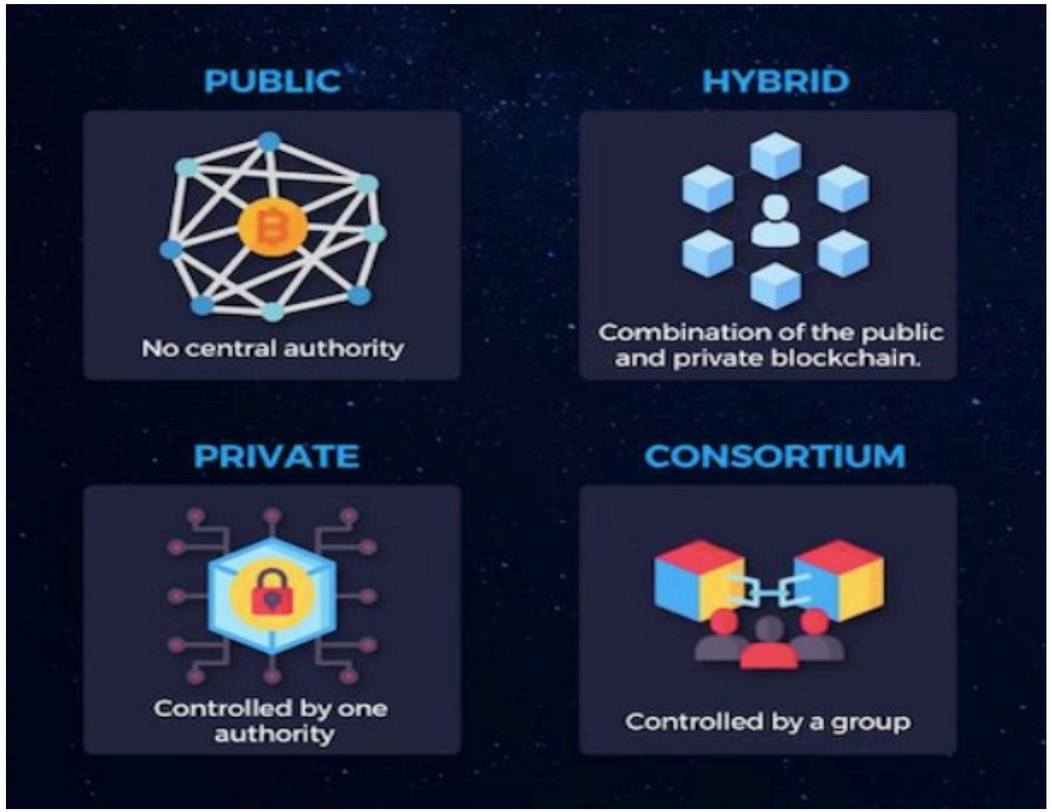
02 Blockchain Architecture

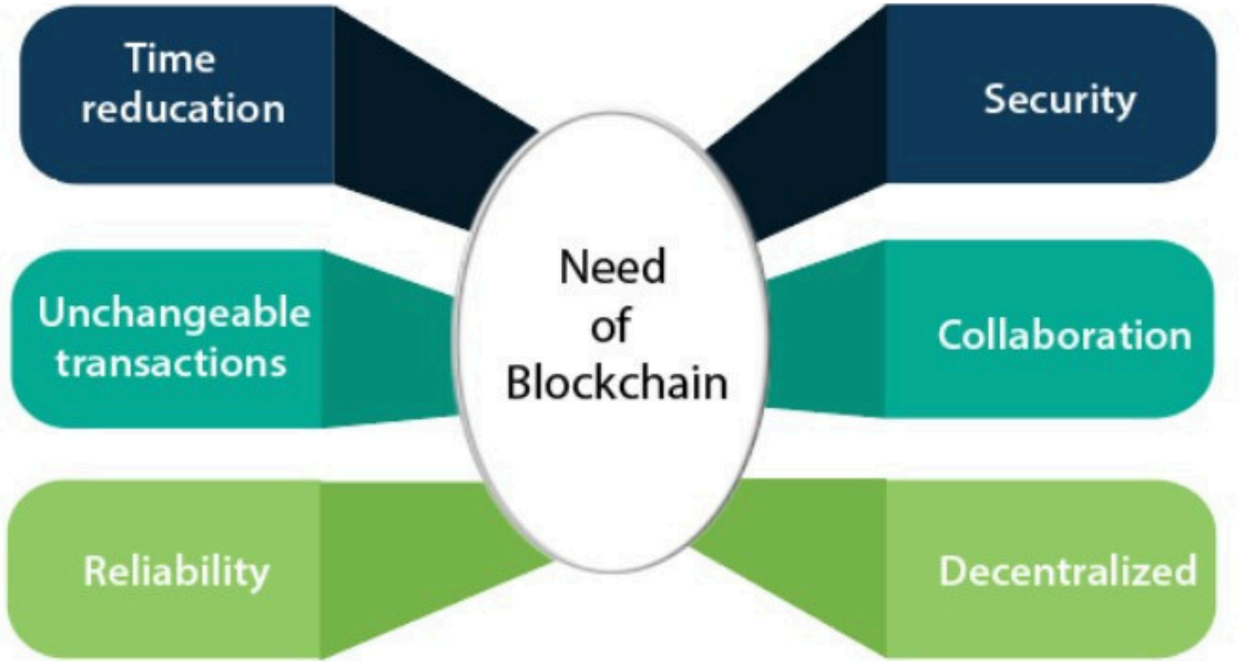
03 Comparative study between existing Blockchains

What is Blockchain?



Blockchain is a decentralized and distributed digital ledger that records transactions across multiple computers or nodes in a transparent and secure manner.





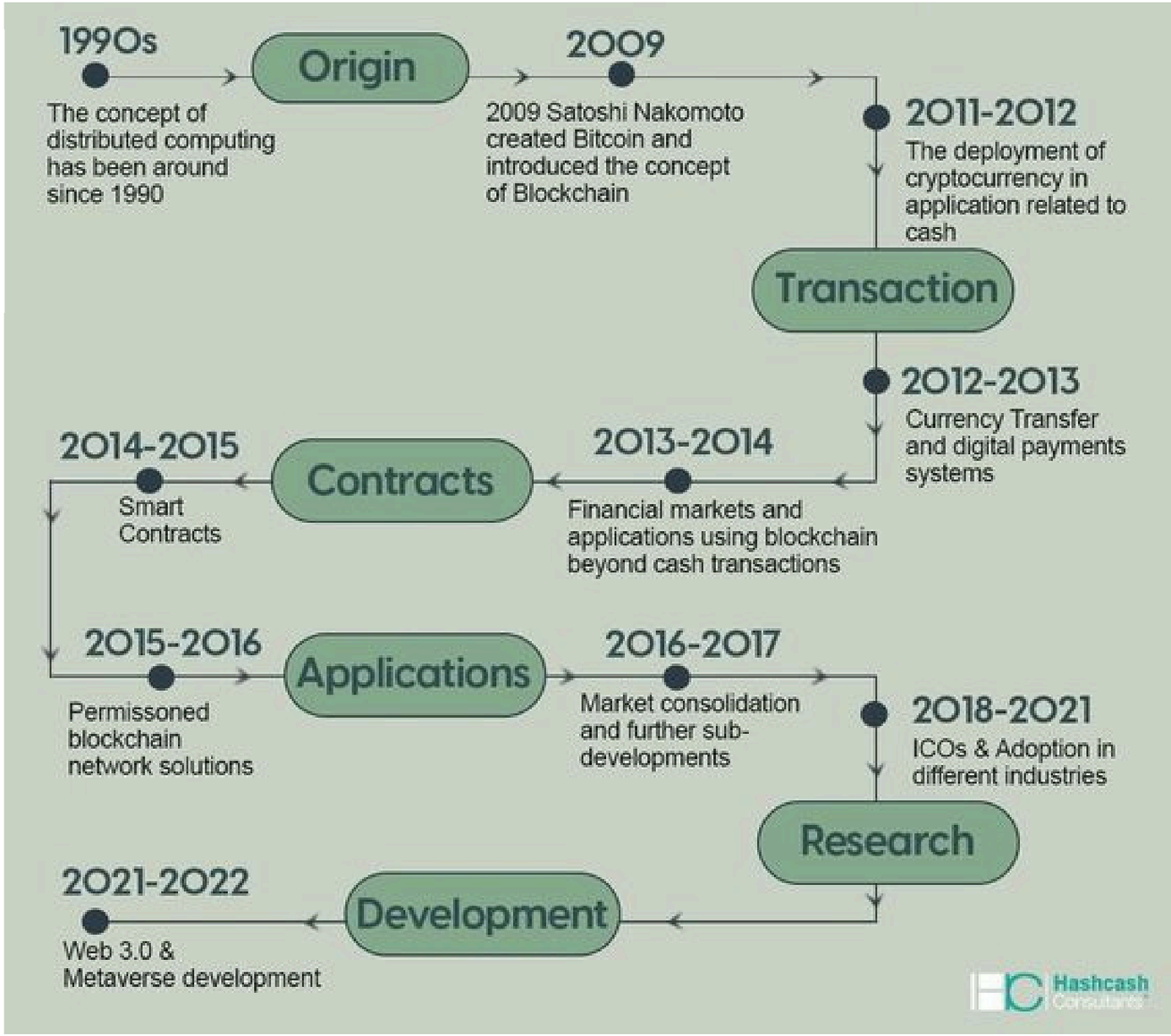
History of Blockchain



Blockchain 3.0: DApps

Blockchain 2.0: Smart Contracts

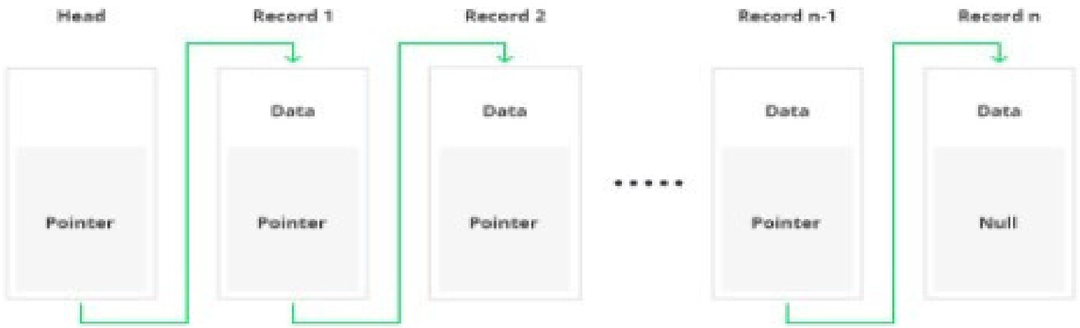
Blockchain 1.0: Currency



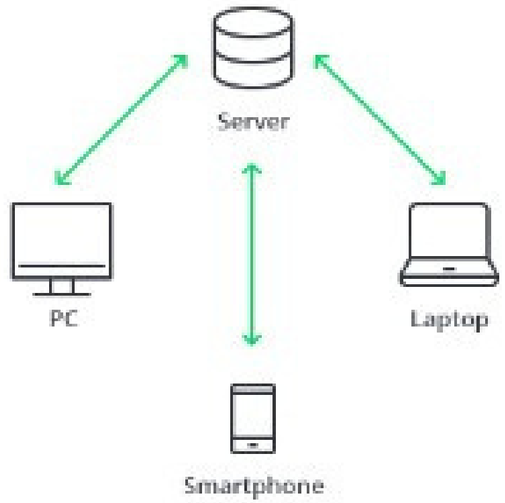
Blockchain Architecture

The blockchain architecture consists of the elements like a node, a data structure used for keeping a set of transactions, and transaction

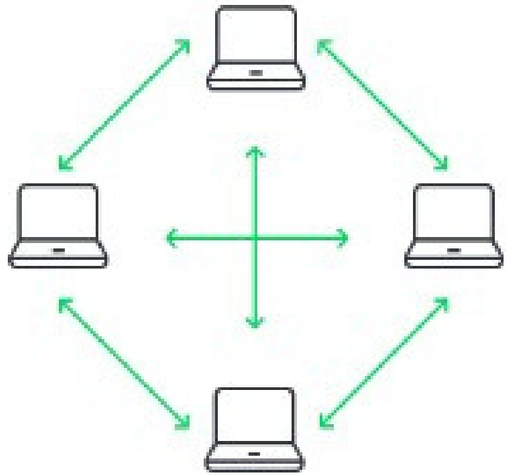
All blocks in a specific order are arranged in a chain with miners that help to verify each block



Database vs. Blockchain Architecture



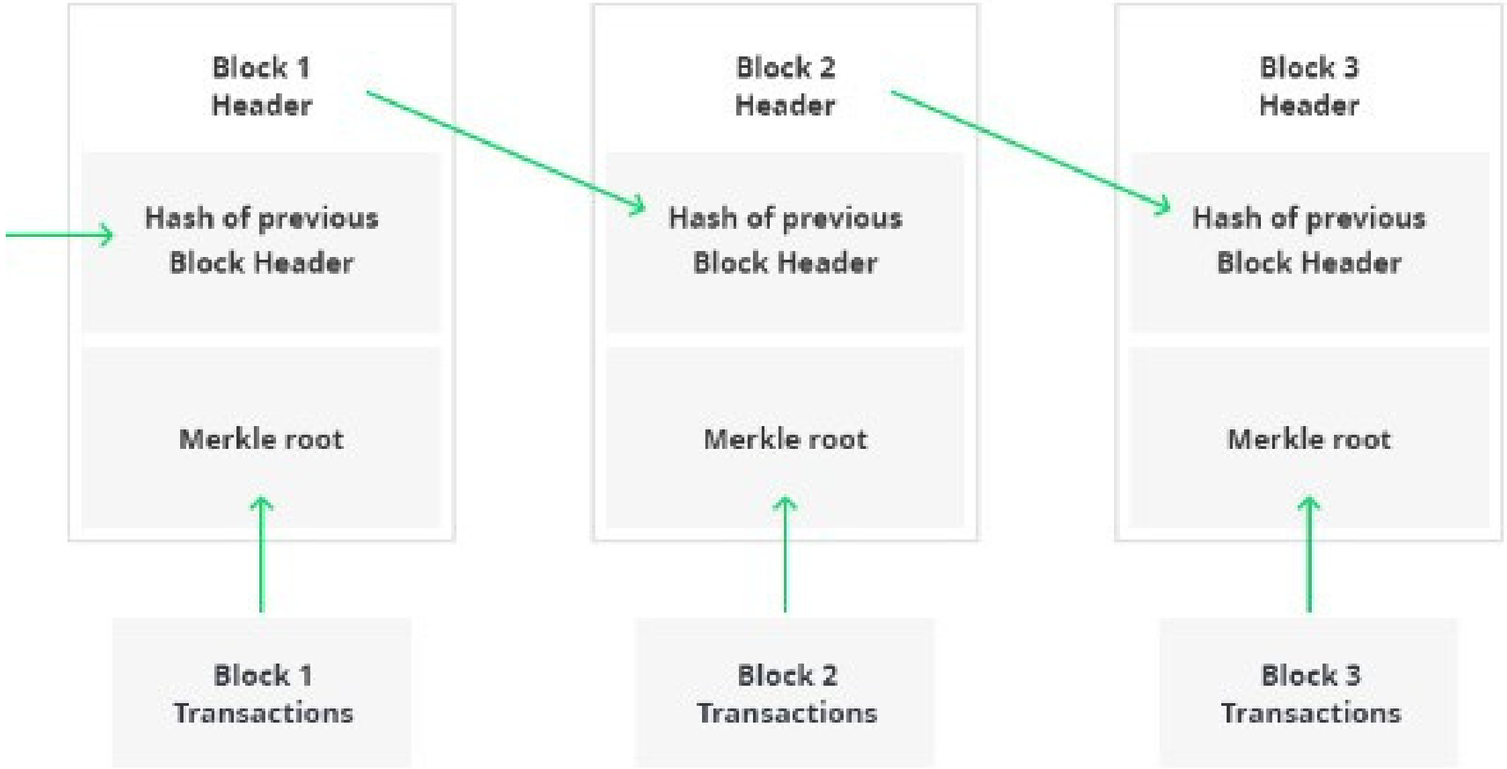
Client-server



P2P network

Blockchain Architecture

Blockchain Hashing



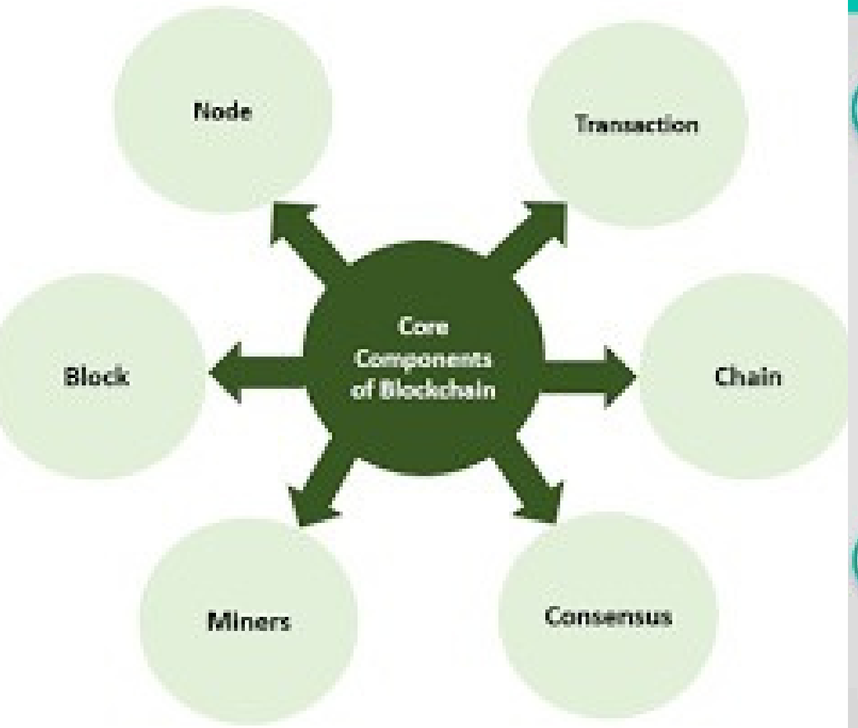
Blockchain Structure

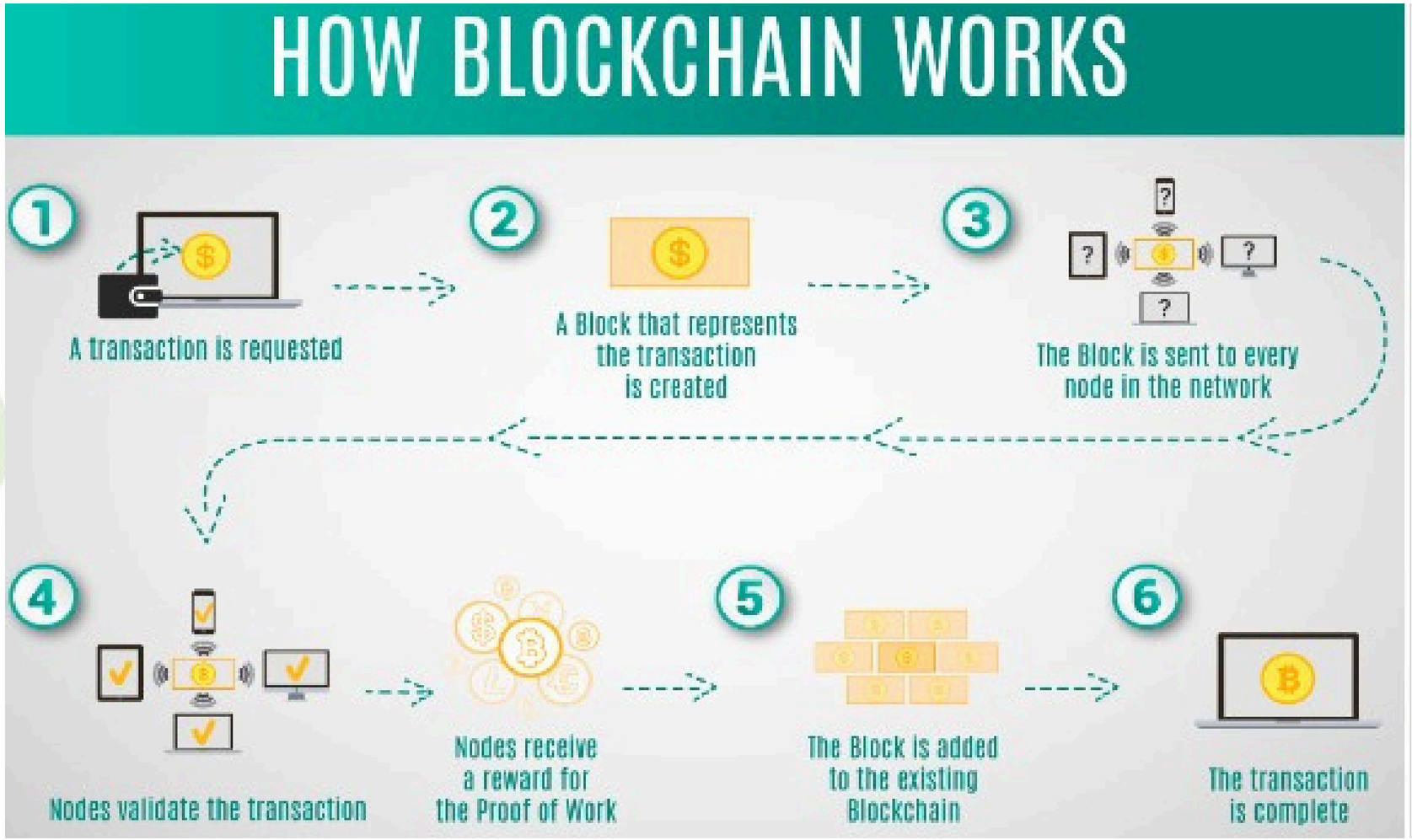
Cost reduction

History of data

Data validity & security

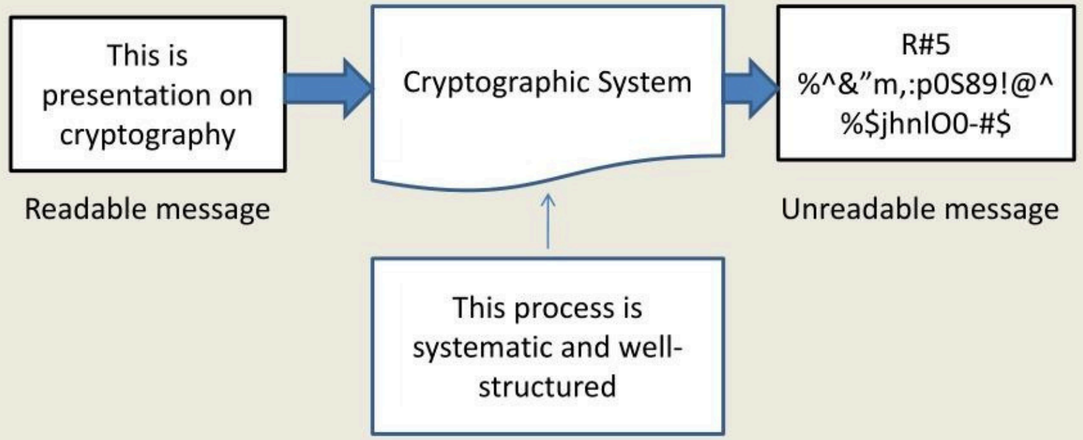
Components of Blockchain Architecture





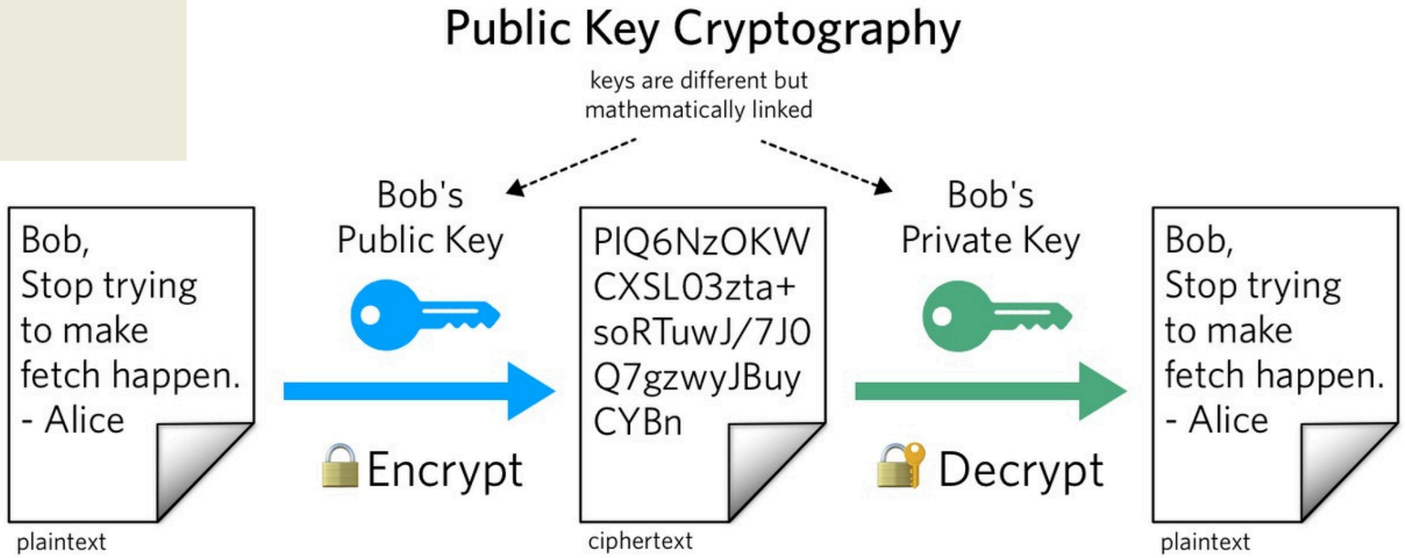
Cryptography

- **Definition :** Cryptography is the art and science of achieving security by encoding message to make them non-readable.



Crypto / Graphy

S ecret / Writing



Framworks - platforms - programming languages



Ethereum

General Blockchain Platform



Corda

Enterprise Blockchain Platform



HyperLedger

Enterprise-Grade
Blockchain Platform



Section 2

Ethereum Blockchain

01 Ethereum

02 Ethereum Ecosystem

03 Ethereum 2.0

– Smart Contracts / DApps

Ethereum

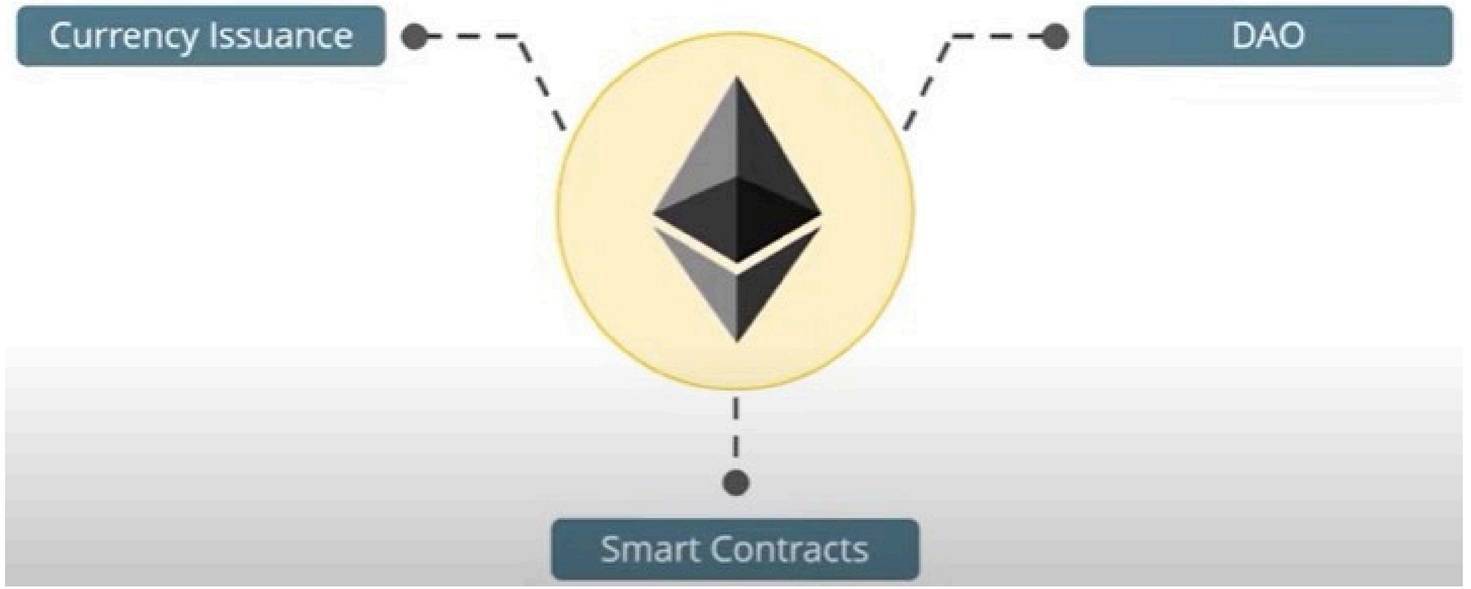
Ethereum is a decentralized platform that runs smart contracts without any downtime, fraud, and third-party interference. And also it has its own crypto currency

Vitalik Buterin: Russian-Canadian computer scientist, entrepreneur, and crypto-philanthropist, co-founder of Ethereum and Bitcoin Magazine .



"The blockchain is not just a glorified database. It is a system for trust without reliance on trust."
- Vitalik Buterin

Ethereum



Features of Ethereum



Ethereum Ecosystem

Gas in Ethereum



It is the crypto fuel for Ethereum. It can be considered a transaction fee which has to be paid when one party sends Ethers to another party.



Ether

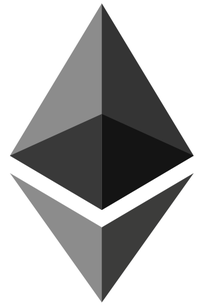


The crypto-fuel of the Ethereum network

Ether is a necessary element needed for operating the Ethereum platform.

Ethereum Ecosystem





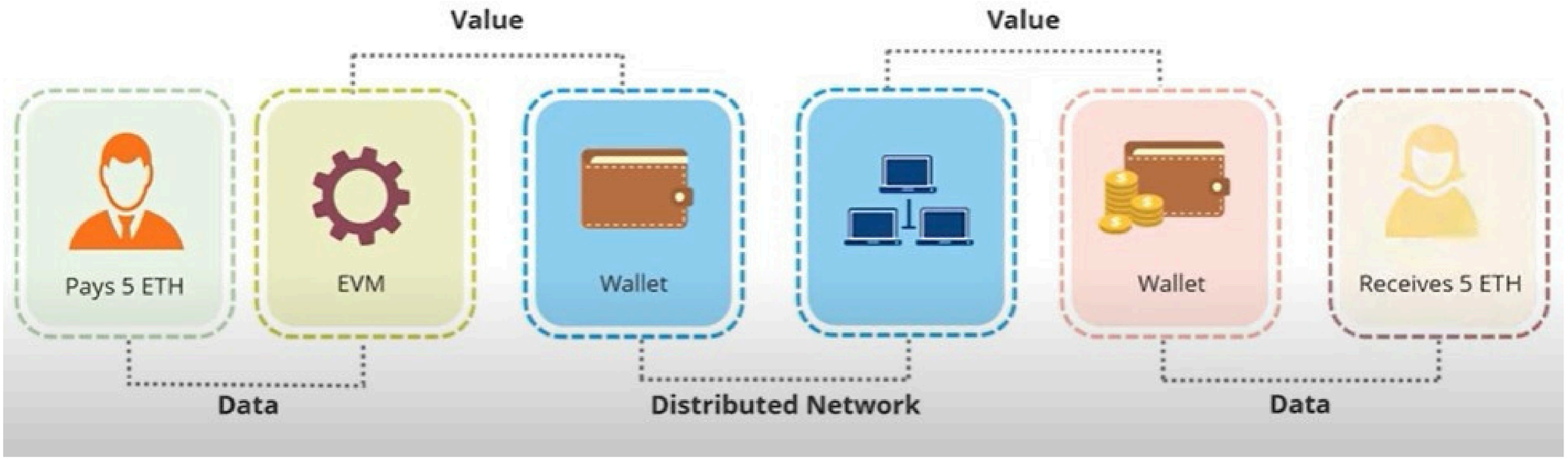
Bitcoin	Ether
Uses secure hash algorithm (SHA-256)	Uses ethash algorithm
Used for purchasing goods and services	Used for making decentralized apps
Is priced at $3500 at the moment	Is priced at $100 at the moment
A currency created to compete against the gold and flat currencies	A token created for facilitating smart contracts
Transaction speed is measured in minutes	Transaction speed is measured in seconds

Ethereum Ecosystem

EVM - Ethereum Virtual Machine

Ethereum Virtual Machine (EVM) is an engine which executes translation code. Smart contracts are compiled into bytecode which an EVM can read and execute.

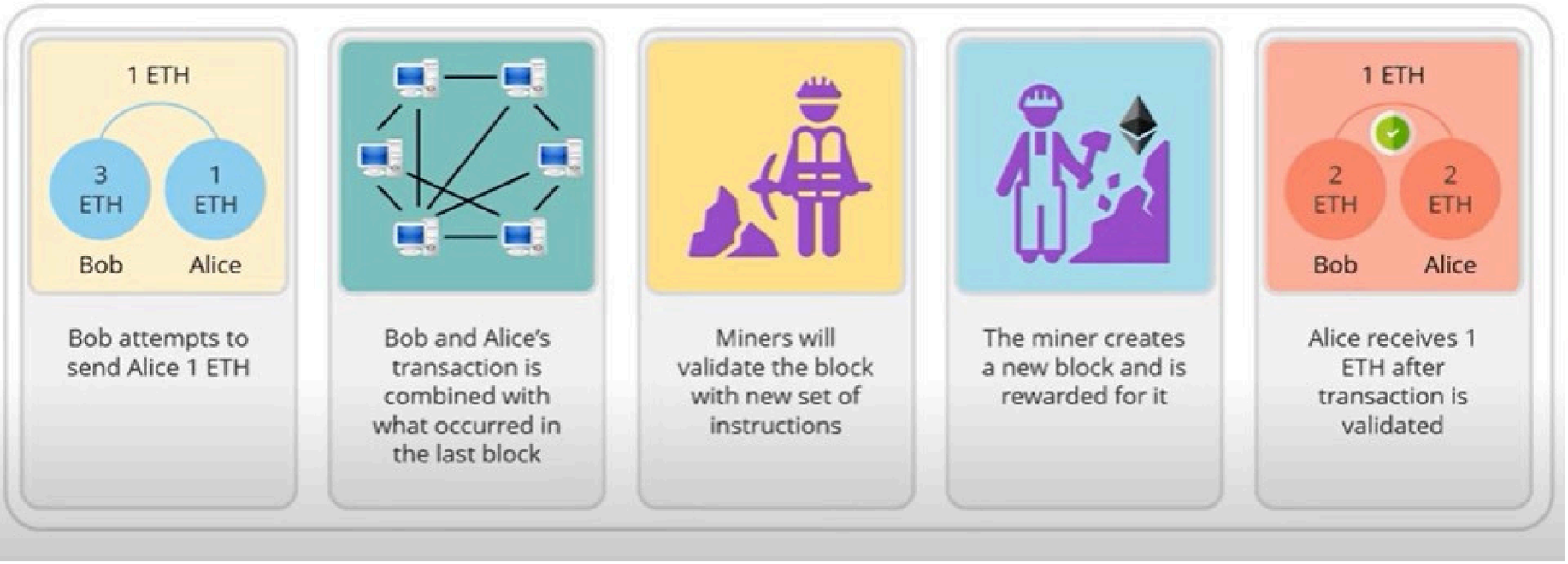
Ethereum possesses its own programming language known as the EVM Bytecode.



Ethereum Ecosystem

Ethereum Mining

Ethereum makes use of Proof of Work mechanism to ensure security. The algorithm used is Ethash.



Ethereum Programming Languages



Ethereum

Smart Contract

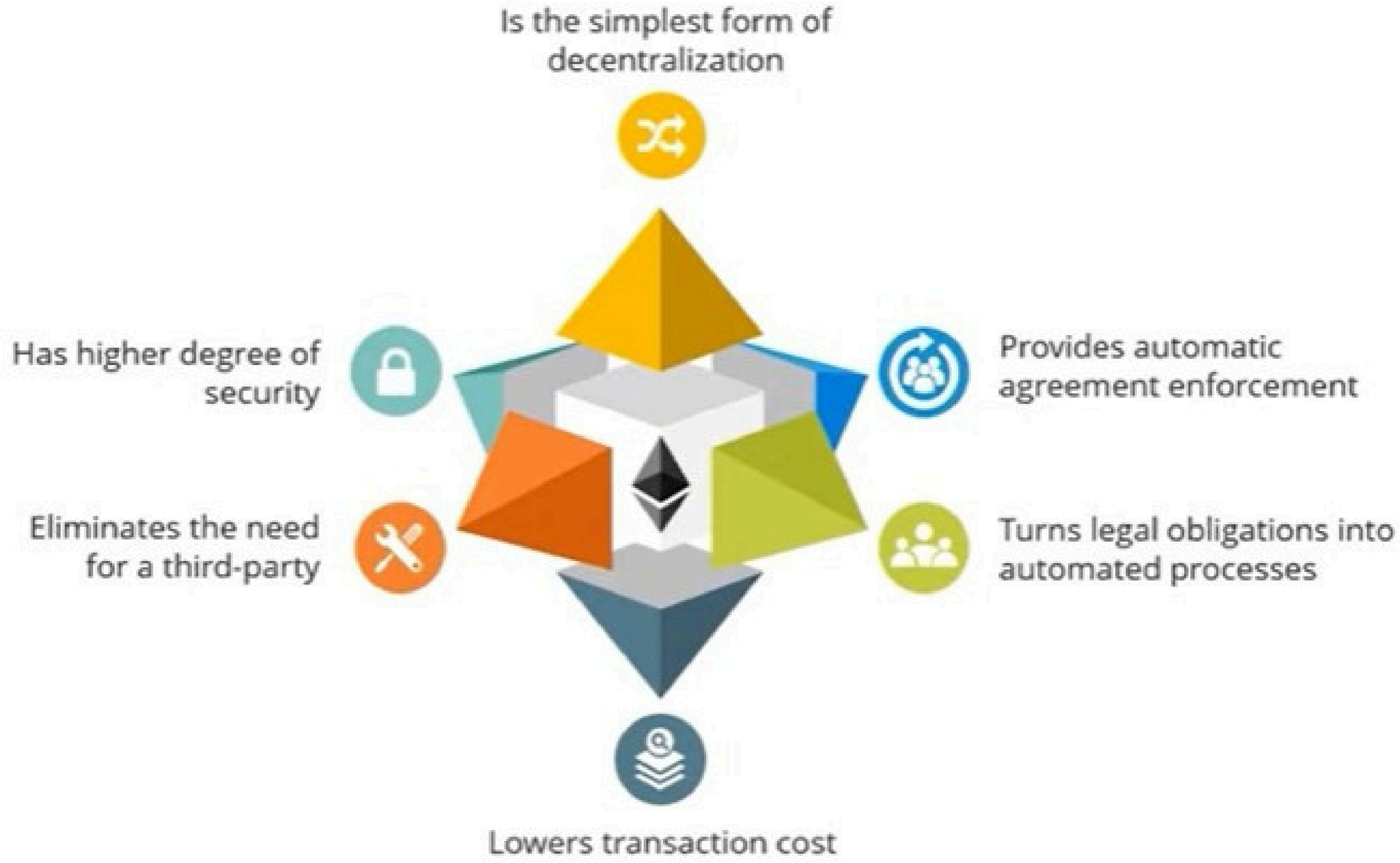
A smart contract is a code running on top of the Blockchain. It contains the set of rules for the nodes to agree upon so that they interact with each other.

1. It has an Ether balance
2. It has an associated code
3. It is triggered by transactions or messages
4. It performs operations of arbitrary complexity



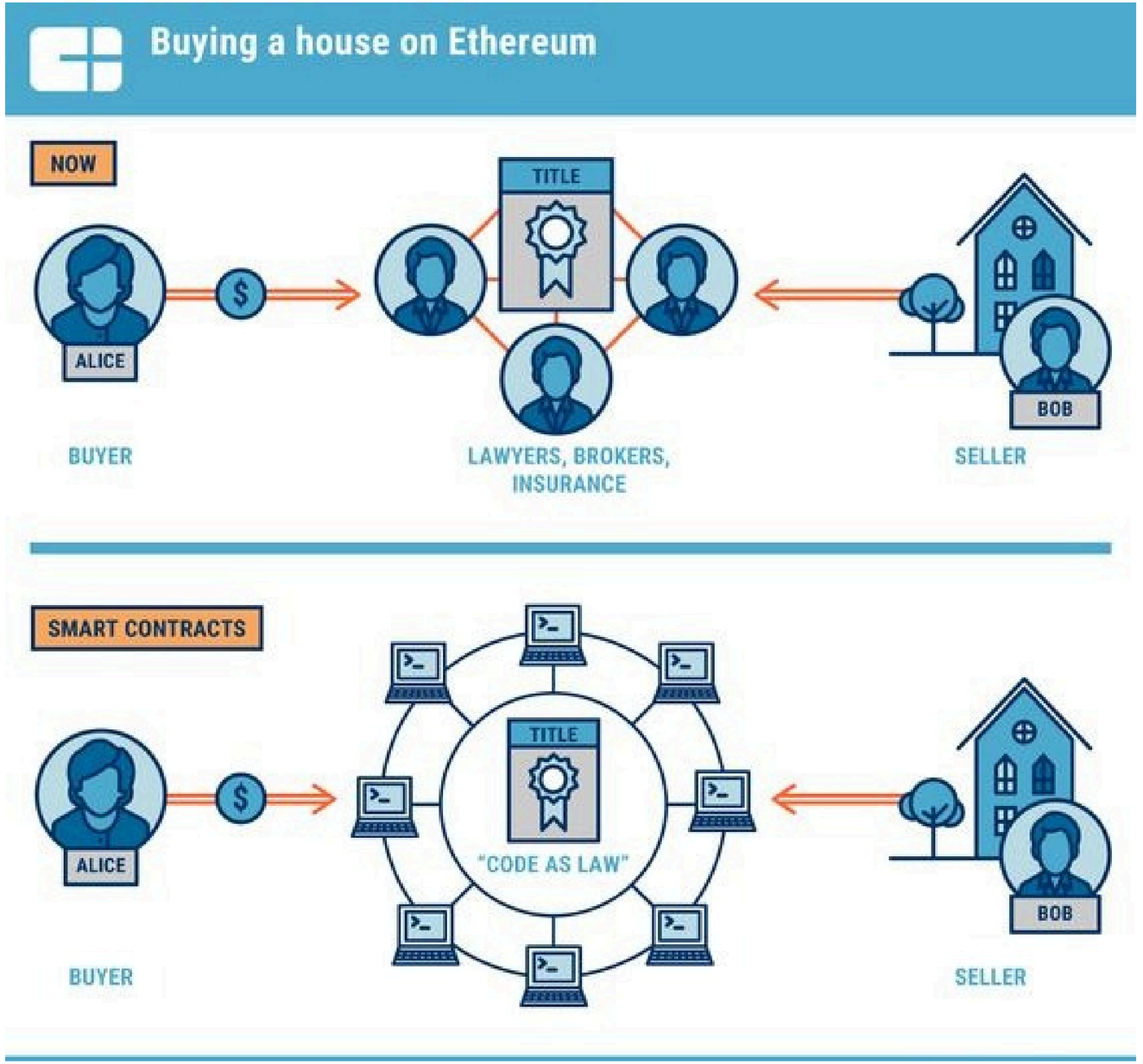
Ethereum

Smart Contract

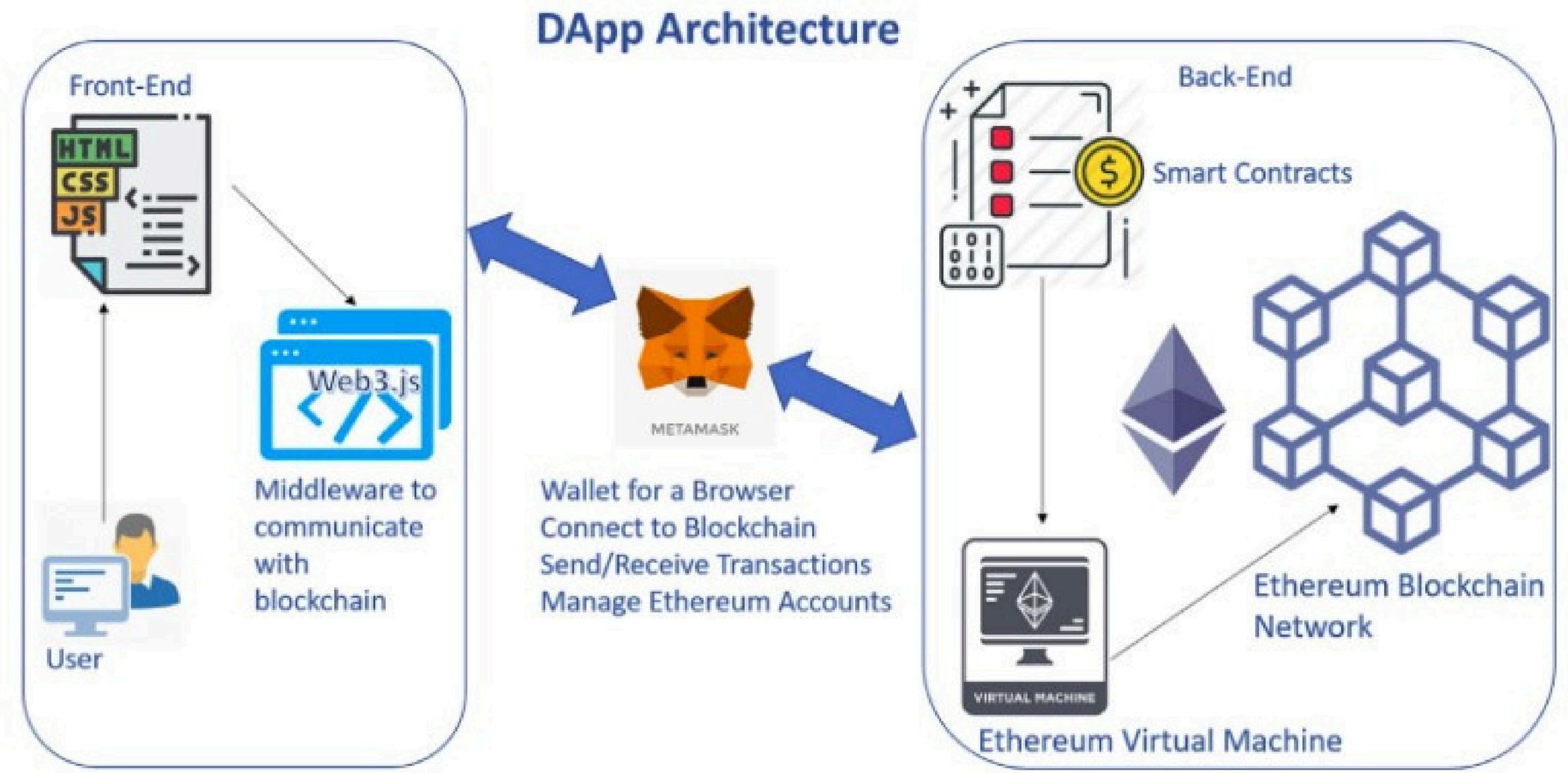


Ethereum

SMART CONTRACT EXAMPLE



DApps


DApp Architecture
Front-End
HTML
CSS
JS
Web3.js
Middleware to communicate with blockchain
User
METAMASK
Wallet for a Browser
Connect to Blockchain
Send/Receive Transactions
Manage Ethereum Accounts
Back-End
Smart Contracts
Ethereum Blockchain Network
VIRTUAL MACHINE
Ethereum Virtual Machine

Section 3

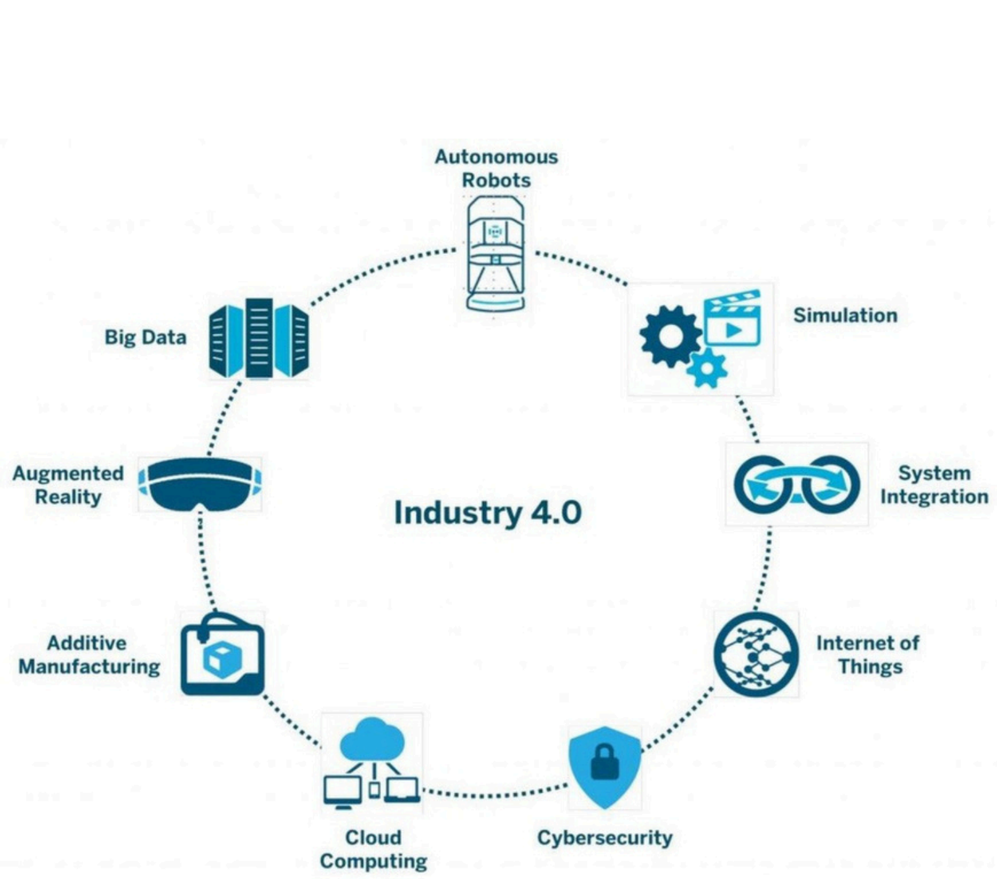
Blockchain in Industry : Supply Chain Management

01 Blockchain && Industry 4.0

02 Use Case: supply chain

03 Supply Chain Management

Blockchain & Industry 4.0





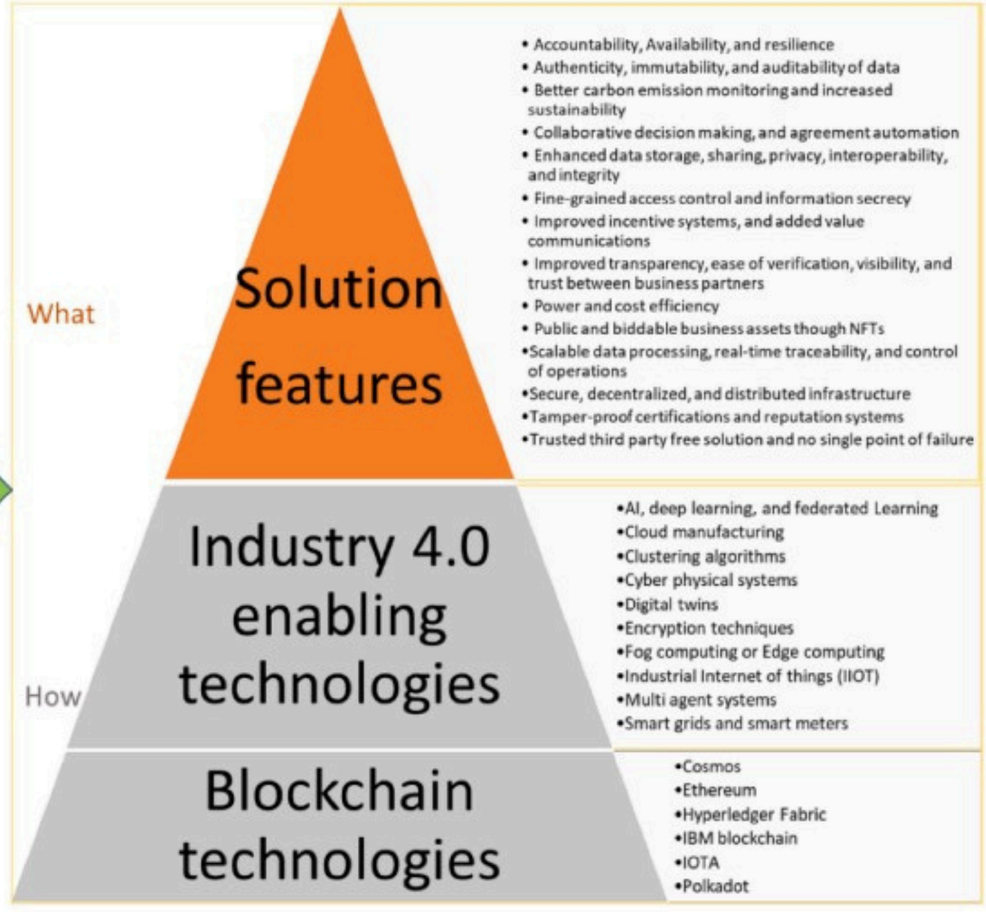
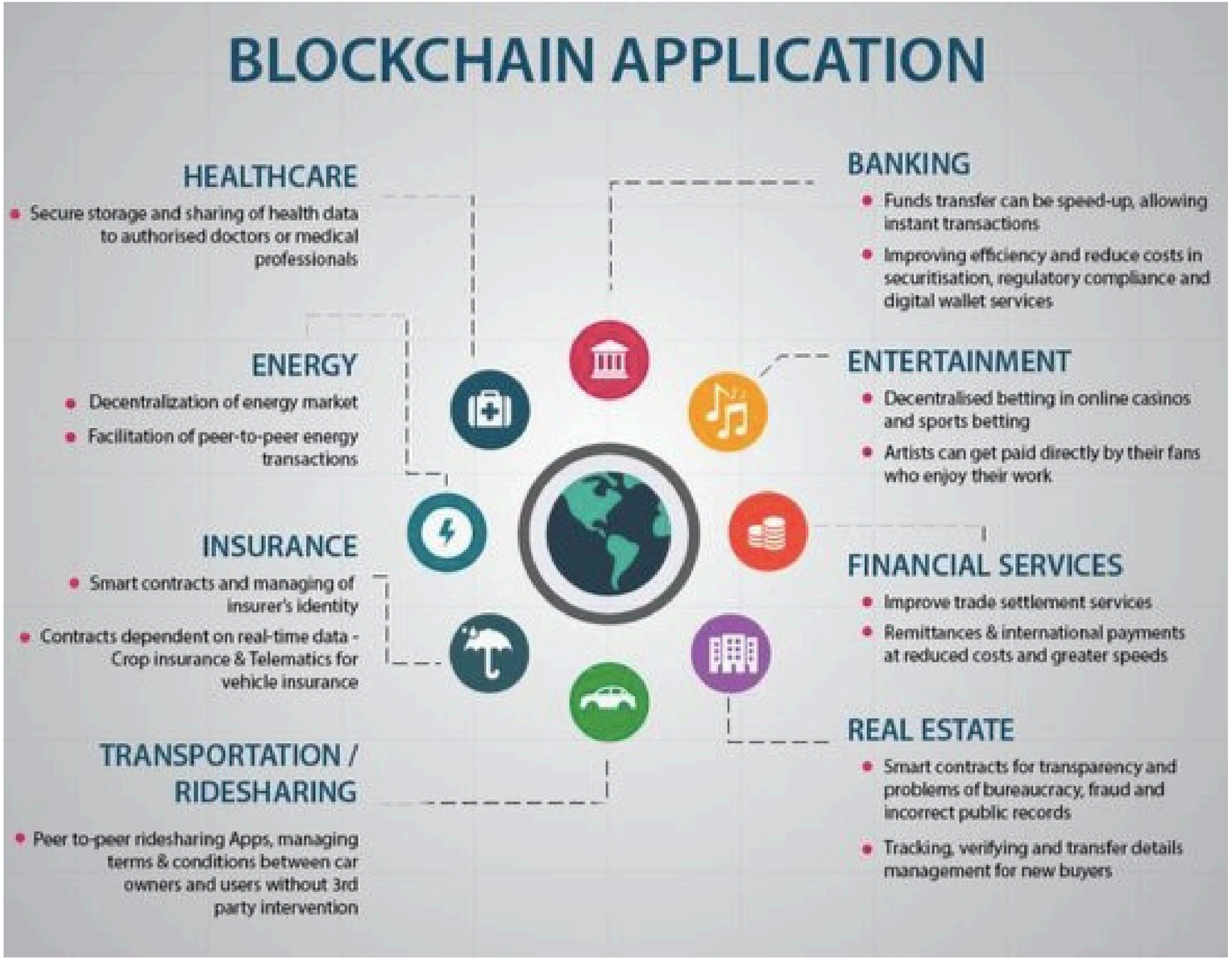


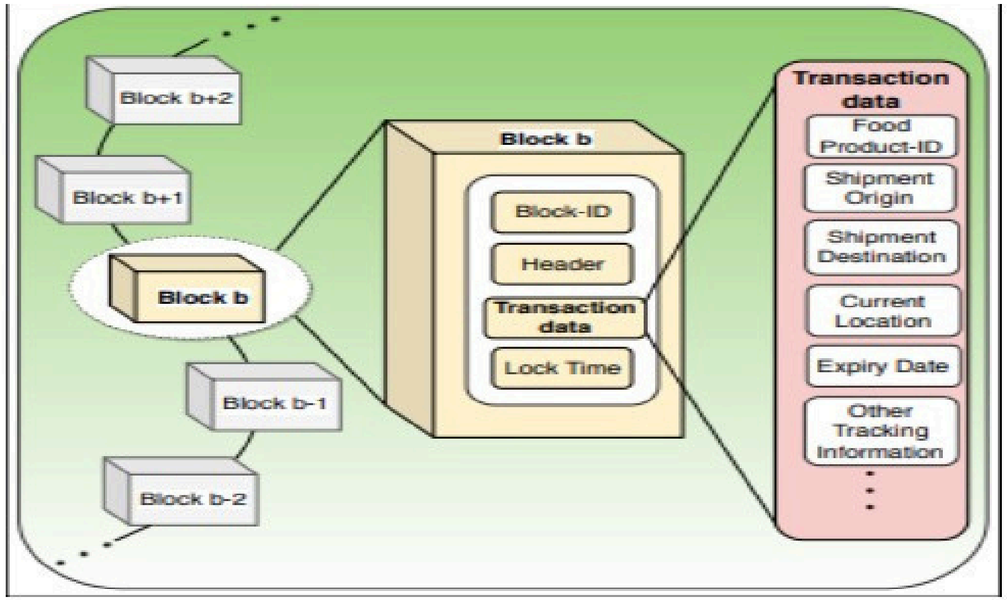
Fig. 2. The blockchain pyramid of applications in Industry 4.0.

Where we can apply Blockchain?



Use Case

Supply Chain 4.0

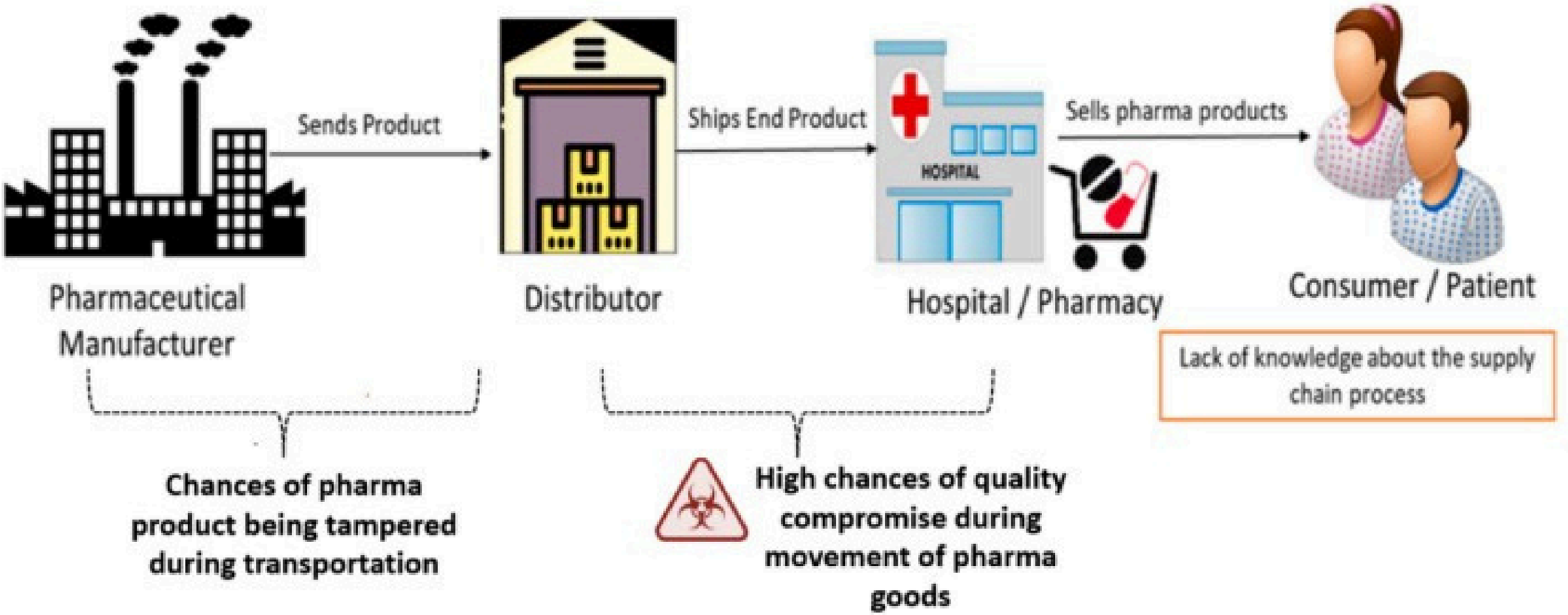


A sample block structure for food product traceability in the supply chain in



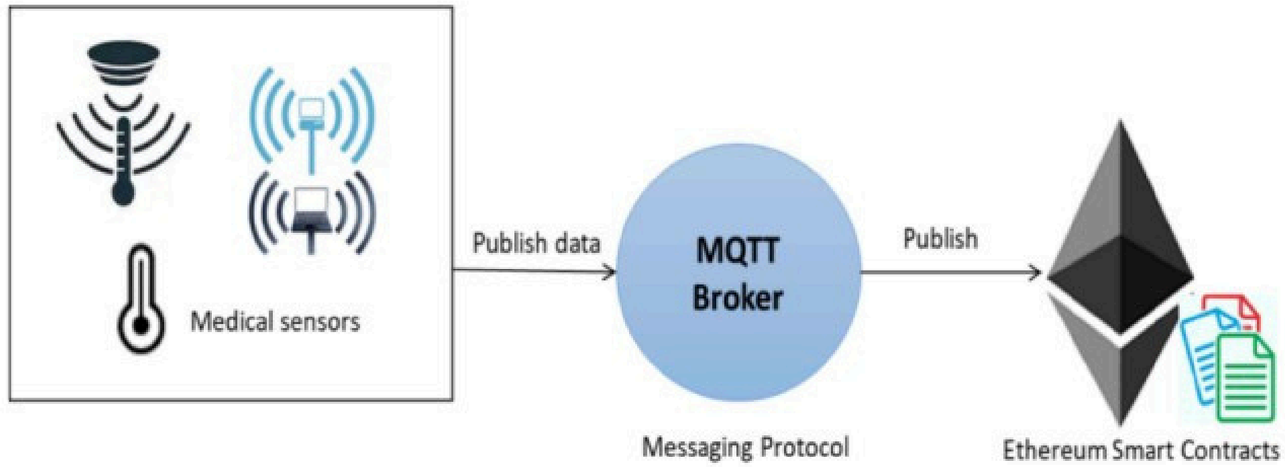
Supply Chain Management

Traditional flow in online Pharma Supply Chain

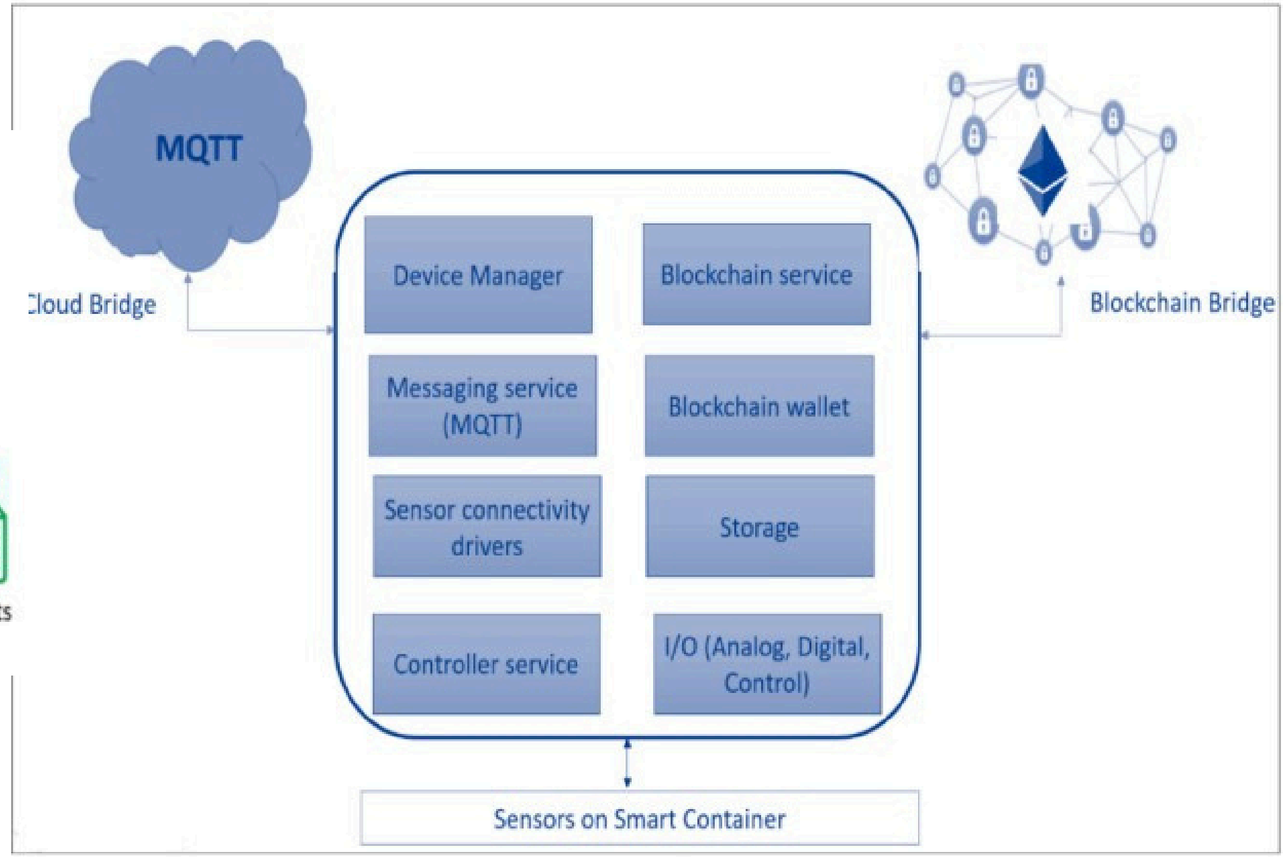


Supply Chain Management

Smart contracts and messaging protocol



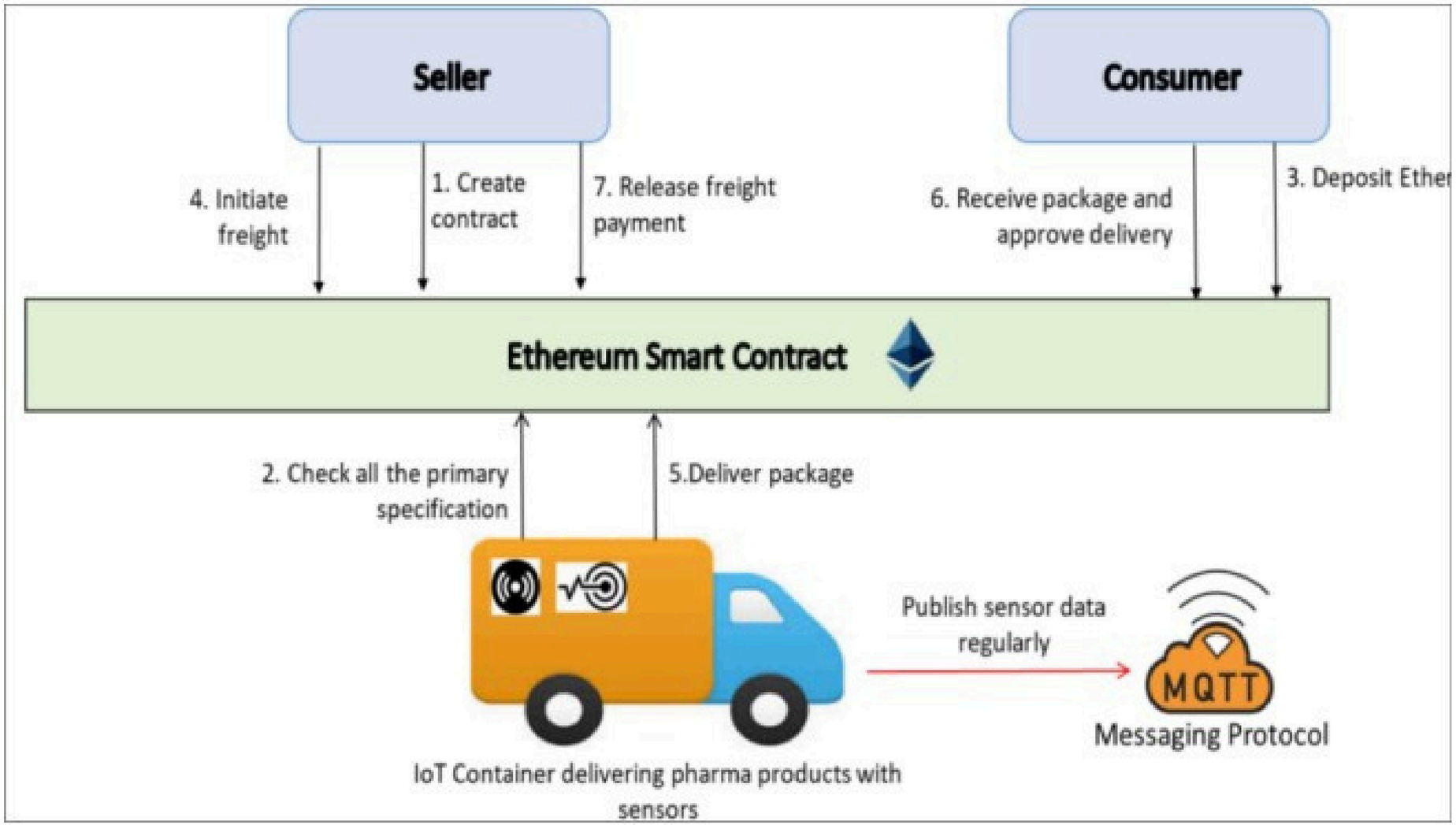
MQTT server for Blockchain-Based



MQTT (Messaging Queuing Telemetry Transport)

Supply Chain Management

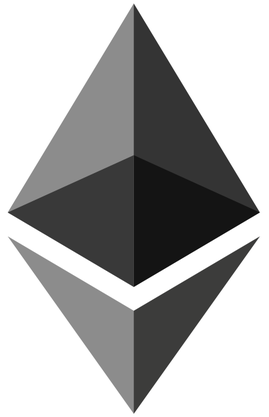
Blockchain Solution for online Pharma Supply Chain



The end

Current Price





31.160,90 USD

1.954,07 USD



Thank You

Walid Hattaoui